

Unit - IV: LINEAR ALGEBRA

Linear Algebra:

Introduction: We have studied the solution of a system of linear equation in one variable x , the solution of a system of linear equations in two variables in x and y . The method of solving a system of linear equations using different methods like Graphical method, Substitution method, Elimination method and cross multiplication method and the nature of solution of the system of linear equations. Linear algebra comprises of matrices and applications of the system of a linear equations. In the present unit, we discuss the rank of the matrix by reducing into row Echelon form, the consistency and inconsistency of the system of a linear equations using the concept of the rank of a matrix. The solution of the system of a linear equations using Gauss - Elimination method. The solution of the system of a linear equations using Gauss - Jacobi's method and Gauss - Seidel method. An eigenvalues and eigenvectors of a square matrix. The eigenvector corresponding to a largest eigenvalue of a matrix using Rayleigh power method. Diagonalization of a square matrix and power of a matrix. The solution of the system of a first order and first degree ordinary differential equations using matrix method.

Continued Unit - IV on Rank of the Matrix

Rank of a Matrix: If A be any given matrix, then the number of non zero rows in a row echelon form of a given matrix A is the rank of the given matrix. The rank of the given matrix A is denoted by $\rho(A)$.

Row echelon form: A non zero matrix A is said to be in row echelon form if it satisfies the following conditions.

1. All the the zero rows are below the non zero rows.
2. The first non zero entry in any non zero row is 1

Note : The row echelon form is nothing but to reducing the given matrix to an upper triangular matrix. The row echelon form generally is in the following form

$$\begin{bmatrix} ** & - & - & \dots & - \\ - & ** & - & \dots & - \\ - & - & ** & \dots & - \\ - & - & - & ** & \dots \end{bmatrix}$$

below the diagonal elements are zero and leading and above the diagonal elements are non zero elements.

Continued Unit - IV on Rank of the Matrix

Examples :

1.

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The given matrix A is in row echelon form and it has 4 non zero rows. The rank of the matrix A is 4 $\therefore \rho(A) = 4$.

2.

$$B = \begin{bmatrix} 1 & 4 & 6 & 3 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The given matrix B is in row echelon form and it has 3 non zero rows. The rank of the matrix B is 3 $\therefore \rho(B) = 3$.

Continued Unit - IV on Rank of the Matrix ...

3.

$$D = \begin{bmatrix} 1 & -5 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The given matrix D is in row echelon form and it has 2 non zero rows. The rank of the matrix D is 2 $\therefore \rho(D) = 2$.

Elementary row Transformations (OR) Elementary row operations:

We are using the following row transformations to reduce the given matrix to row echelon form

- ❶ Interchanging any two rows of the matrix and it is denoted by

$$R_1 \longleftrightarrow R_2, R_2 \longleftrightarrow R_3$$

- ❷ Any row is multiple by a non zero scalar constant k and it is denoted by

$$R_2 \rightarrow kR_1, R_3 \rightarrow -kR_2$$

- ❸ Adding or Subtracting any row a non zero scalar constant multiple of any other row and it is denoted by

$$R_2 \rightarrow R_2 + kR_1, R_3 \rightarrow R_3 - kR_1$$

Continued Unit - IV Problems on Rank of the Matrix ...

Problems:

Problem - 01: Find the rank of the matrix by reducing to row echelon form.

$$\begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}$$

Solution: Take

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}$$

Applying elementary row transformations reducing the given matrix to row echelon form

$$R_2 \longrightarrow R_2 - 2R_1 \text{ and } R_3 \longrightarrow R_3 - R_1$$

$$A \approx \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix}$$

Continued Unit - IV Problems on Rank of the Matrix ...

$$R_3 \longrightarrow R_3 - 2R_2$$

$$A \approx \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

Therefore, the given matrix A is in row echelon form and it has 2 non zero rows. The rank of the matrix A is 2. $\therefore \rho(A) = 2$.

Problem - 02: Find the rank of the matrix by reducing to row echelon form.

$$\begin{bmatrix} 2 & -1 & 1 \\ 3 & -1 & 1 \\ 4 & -1 & 2 \\ -1 & 1 & -1 \end{bmatrix}$$

Solution: Take

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 3 & -1 & 1 \\ 4 & -1 & 2 \\ -1 & 1 & -1 \end{bmatrix}$$

Continued Unit - IV Problems on Rank of the Matrix ...

Applying elementary row transformations to the given matrix and reducing to row echelon form

$$R_2 \longrightarrow 2R_2 - 3R_1,$$

$$R_3 \longrightarrow R_3 - 2R_1$$

$$R_4 \longrightarrow 2R_4 + R_1$$

$$A \approx \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$$R_3 \longrightarrow R_3 - R_2$$

$$R_4 \longrightarrow R_4 - R_2$$

$$A \approx \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Therefore, the given matrix A is reduced to row echelon form. The number of non zero rows are 3 and hence the rank of the given matrix A is 3. $\therefore \rho(A) = 3$.

Continued Unit - IV Problems on Rank of a Matrix ...

Problem - 03: Find the rank of the matrix by reducing to row echelon form

$$\begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ -3 & 1 & 2 \end{bmatrix}$$

Solution: Take

$$A = \begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ -3 & 1 & 2 \end{bmatrix}$$

Reducing the given matrix to row echelon form by applying elementary row transformations

$$R_2 \longrightarrow R_2 + 2R_1$$

$$R_3 \longrightarrow R_3 + R_1$$

$$A \approx \begin{bmatrix} 3 & -1 & 2 \\ 0 & 0 & 8 \\ 0 & 0 & 4 \end{bmatrix}$$

Continued Unit - IV Problems on Rank of a Matrix ...

$$R_3 \longrightarrow 2R_3 - R_2$$

$$A \approx \begin{bmatrix} 3 & -1 & 2 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

Now, it is in row echelon form and has two non zero rows and hence the rank of the given matrix A is 2.

$$\therefore \rho(A) = 2.$$

Problem - 04: Find the rank of the matrix by reducing to row echelon form

$$\begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$$

Solution: Take

$$B = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$$

Continued Unit - IV Problems on Rank of a Matrix ...

Reducing the given matrix to row echelon form by applying elementary row transformations

$$R_1 \longleftrightarrow R_2$$

$$B \approx \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$$

$$R_3 \longrightarrow R_3 - 2R_1$$

$$B \approx \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \end{bmatrix}$$

$$R_3 \longrightarrow R_3 - R_2$$

$$B \approx \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now, it is in row echelon form and has two non zero rows and hence the rank of the given matrix B is 2.

$$\therefore \rho(B) = 2.$$

Continued Unit - IV Problems on Rank of a Matrix ...

Problem - 05: Find the rank of the matrix by reducing to row echelon form

$$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Solution: Take

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Reducing the given matrix to row echelon form by applying elementary row transformations

$$R_1 \longleftrightarrow R_2$$

$$A \approx \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Continued Unit - IV Problems on Rank of a Matrix ...

$$R_3 \longrightarrow R_3 - 3R_1$$

$$R_4 \longrightarrow R_4 - R_1$$

$$A \approx \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix}$$

$$R_3 \longrightarrow R_3 - R_2$$

$$R_4 \longrightarrow R_4 - R_2$$

$$A \approx \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now, it is in row echelon form and has two non zero rows and hence the rank of the given matrix A is 2.

$$\therefore \rho(A) = 2.$$

Continued Unit - IV Problems on Rank of a Matrix ...

Problems: Find the rank of the following matrices, by elementary row transformations.

$$\begin{aligned} (1). & \begin{bmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 7 & 22 \end{bmatrix} \quad (2). \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix} \quad (3). \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix} \quad (4). \begin{bmatrix} 1 & 1 & 1 & -3 \\ 3 & 1 & -2 & -2 \\ 2 & 4 & 7 & 7 \end{bmatrix} \\ (5). & \begin{bmatrix} 1 & 3 & 2 \\ 2 & -1 & 3 \\ 3 & -5 & 4 \\ 1 & 17 & 4 \end{bmatrix} \quad (6). \begin{bmatrix} 3 & 4 & -1 & -6 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 1 & 3 & 13 & 3 \end{bmatrix} \quad (7). \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 1 & 2 \end{bmatrix} \\ (8). & \begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & -1 & 2 \end{bmatrix} \end{aligned}$$

Continued Unit - IV on Consistent of a system of a Linear Equations

Consistency of system of a linear equations:

A system of equations are said to be linear in which all unknown quantities (or unknown variables) $x_1, x_2, x_3, \dots, x_n$ are appear in first degree alone.

Consider a set of m system of a linear equations in n unknowns quantities (or unknown variables) $x_1, x_2, x_3, \dots, x_n$ as follows.

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

...

...

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

where a_{mn} 's, $m = 1, 2, 3, \dots, n = 1, 2, 3, \dots$ and $b_1, b_2, \dots, b_m$ are constants.

system of a Homogeneous linear equations: If the RHS quantities $b_1, b_2, \dots, b_m$ all are equal to zero then the above system of a linear equations are said to be a **system of a homogeneous linear equations**.

Non - Homogeneous system of a linear equations: A system of a linear equations are said to be a **Non homogeneous system of a linear equations** in which the RHS quantities $b_1, b_2, \dots, b_m$ are not equal to zero.

Continued Unit - IV on Consistent of a system of a Linear Equations ...

Solution: The set unknown quantities $x_1, x_2, x_3, \dots, x_n$ are satisfies the system of a linear equations is called as a **solution** of the system of a linear equations.

Consistency: A system of a linear equations are said to be **consistency** if it has a solution. A system of a linear equations is consistent and has two types of solutions, they are (a) an unique (OR) (b) an infinitely many solutions.

Inconsistency: A system of linear equations are said to be **Inconsistency** if it has no solution.

Trivial Solution : If all the unknown quantities $x_1, x_2, x_3, \dots, x_n$ are all zero then the solution is called as **trivial solution** and it is a solutions for a system of homogeneous linear equations.

Non trivial solution : If at least one unknown quantity is not equal to zero is called as **non trivial solution**.

Therefore, the above system of a linear equations can be representing in matrix equation of the form

$$AX = B$$

Where A is called the coefficient matrix, X is a column matrix of all the unknown quantities $x_1, x_2, x_3, \dots, x_n$ and B is also a column matrix of the RHS values of the above system of a linear equations

Continued Unit - IV on Consistent of a system of a Linear Equations ...

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \dots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \dots \\ \dots \\ b_m \end{bmatrix}$$

and the augmented matrix of the above system of linear equations is of the form $[A : B]$ and it is given by

$$[A : B] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} & : & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} & : & b_2 \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} & : & b_3 \\ \dots & \dots & \dots & \dots & \dots & : & \dots \\ \dots & \dots & \dots & \dots & \dots & : & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} & : & b_m \end{bmatrix}$$

Continued Unit - IV on Consistent of a system of Linear Equations ...

Note: The system of a linear equations $AX = B$ may or may not possess a solution. Suppose the system of a linear equations possesses a solution it may or may not be the only solution. The concept of the rank of the matrix is useful us to conclude the system of a linear equation is consistent or not and the system of a linear equations has an unique solution or an infinitely many solutions.

Conditions for Consistent and Inconsistent :

Consistent: A system of a linear equation are said to be consistent if

$\rho(A) = \rho(A: B)$ and it has two types of solutions, they are,

(i) **Consistent with an unique solution:** A system of a linear equation are said to be consistent and has an unique solution if $\rho(A) = \rho(A: B) = r = n$.

where r is the rank and n is the number of unknowns in the given system of linear equations.

(ii) **Consistent with an infinitely many solutions:** A system of a linear equation are said to be consistent and has an infinitely many solutions if $\rho(A) = \rho(A: B) = r < n$. here we have assume that $(n - r)$ unknowns are arbitrary constants.

Inconsistent or No solution: A system of a linear equation are said to be inconsistent or no solution if $\rho(A) \neq \rho(A: B)$

Geometrical Representation of a system of a linear equations for consistent and Inconsistent:

Consistent and has an unique solution: A system of a linear equations in two variables x and y are consistent and has an unique solution then geometrical represents that the straight lines which are intersects at one point in the xy plane.

Example: Consider a system of a linear equations in two variables x and y are $x + 2y = 3$, $4x + 5y = 6$, the point of intersection is $(-1, 2)$.

Consistent and has an infinitely many solutions: A system of a linear equations in two variables x and y are consistent and has an infinitely many solutions then geometrical represents that the straight line are intersects at many points or coincide each other in the xy plane.

Example: Consider a system of a linear equations in two variables x and y are $x + 2y = 3$, $4x + 8y = 12$, the points of intersection are more than one point.

Inconsistent or No solution: A system of the linear equations in two variables x and y are inconsistent or no solution then geometrical represents that the straight lines are not intersects at any point or parallel to each other in the xy plane.

Example: Consider a system of the linear equations in two variables x and y are $x + 2y = 3$, $4x + 8y = 6$, there is no point of intersection.

Continued Unit - IV on Consistent of a system of Linear Equations ...

Note - 01: A system of a homogeneous linear equations is always consistent because it has at least one trivial solution.

Note - 02: A system of a homogeneous linear equations has a non - trivial solution if and only if the determinant of the coefficient matrix of the system of homogeneous linear equations is zero. that is $|A| = 0$

Note - 03: A system of a non homogeneous linear equations may or may not be consistent.

Working procedure:

Step - 01: Write the given system of a linear equations in matrix equation $AX = B$.

Step - 02: Write the augmented matrix of the form $[A : B]$.

Step - 03: Applying the series elementary row transformations to augmented matrix and reduce the augmented matrix $[A : B]$ to row echelon form.

Step - 04: Find the rank of the coefficient matrix A and augmented matrix $[A : B]$. Verify the conditions for consistent and inconsistent of the system of equations.

Step - 05: If it is consistent solve the unknown quantities of the given system of a linear equations.

Note: If the given system of a linear equations are consistent and has infinitely many solutions, we have to assume that $(n - r)$ unknowns are arbitrary constants.

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

Problems:

Problem - 01: Find the value of k for which the system of linear equation $(3k - 8)x + 3y + 3z = 0$, $3x + (3k - 8)y + 3z = 0$, $3x + 3y + (3k - 8)z = 0$ has a non-trivial solution.

Solution: Given the homogeneous system of a linear equations are

$$(3k - 8)x + 3y + 3z = 0, 3x + (3k - 8)y + 3z = 0, 3x + 3y + (3k - 8)z = 0$$

We know that, the homogeneous system of a linear equations has non trivial solution if and if the determinant of the coefficient matrix is equal to zero, *i.e.* $|A| = 0$

$$|A| = 0$$

$$\begin{vmatrix} 3k - 8 & 3 & 3 \\ 3 & 3k - 8 & 3 \\ 3 & 3 & 3k - 8 \end{vmatrix} = 0$$

Now, expanding the determinant, we get

$$[(3k - 8)(3k - 8)^2 - 9] - 3[3(3k - 8) - 9] + 3[9 - 3(3k - 8)] = 0$$

$$(3k - 8)(3k - 11)(3k - 5) - 18(3k - 11) = 0$$

$$(3k - 11)^2(3k - 2) = 0$$

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

From this we get, $k = \frac{11}{3}$ or $k = \frac{2}{3}$

Hence the given homogeneous system of linear equations has a non - trivial solution for $k = \frac{11}{3}$ or $k = \frac{2}{3}$.

Problem - 02: Test for consistency and solve the system of linear equation $x + y + z = 4$, $2x + y - z = 1$, $x - y + 2z = 2$.

Solution: Given the system of linear equations

$$x + y + z = 4, 2x + y - z = 1, x - y + 2z = 2$$

The matrix equation of the given system of linear equations is

$$AX = B$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

The augmented matrix of the given system of linear equations is

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 1 & -1 & 2 & 2 \end{array} \right]$$

Reducing the augmented matrix $[A : B]$ to row echelon form by applying the elementary row transformations.

$$R_2 \longrightarrow R_2 - 2R_1$$

$$R_3 \longrightarrow R_3 - R_1$$

$$[A : B] \approx \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -1 & -3 & -7 \\ 0 & -2 & 1 & -2 \end{array} \right]$$

$$R_3 \longrightarrow R_3 - 2R_2$$

$$[A : B] \approx \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -1 & -3 & -7 \\ 0 & 0 & 7 & -12 \end{array} \right]$$

Continued Unit - IV on Consistent of a system of Linear Equations ...

Therefore, the augmented matrix $[A; B]$ is in row echelon and $\rho(A) = 3$, $\rho(A : B) = 3$
 $\therefore \rho(A : B) = \rho(A) = 3 = r = (n = 3)$. Hence the given system of a linear equations are consistent and has an unique solution.

$$x + y + z = 4 \quad (1)$$

$$-y - 3z = -7 \quad (2)$$

$$7z = -12 \quad (3)$$

Now, solve by back substitution method, from the Eqn. (3) $z = \frac{-12}{7}$, from Eqn. (2)
 $-y = -7 + 3z$, $y = \frac{-85}{7}$ and from the Eqn. $x = 4 - y - z$, $x = \frac{125}{7}$
Hence, $x = \frac{125}{7}$, $\frac{-85}{7}$ and $z = \frac{-12}{7}$ are the unique solution for the given system of a linear equations.

Continued Unit - IV Problems on Consistent of a system of a linear equations ...

Problem - 02: Test consistency and solve $5x + 3y + 7z = 4$, $3x + 26y + 2z = 9$, $7x + 2y + 10z = 5$

Solution: Given the system of linear equations

$$5x + 3y + 7z = 4, 3x + 26y + 2z = 9, 7x + 2y + 10z = 5$$

The given system of a linear equations are in matrix equation

$$AX = B$$
$$\begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix}$$

The augmented matrix of the given system of linear equations $[A : B]$ is

Continued Unit - IV Problems on Consistent of a system of a linear equations ...

$$[A : B] = \left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 3 & 26 & 2 & 9 \\ 7 & 2 & 10 & 5 \end{array} \right]$$

Reducing the augmented matrix $[A : B]$ to row echelon form by applying the elementary row transformations.

$$R_2 \longrightarrow 5R_2 - 3R_1$$

$$R_3 \longrightarrow 5R_3 - 7R_1$$

$$[A : B] \approx \left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 0 & -11 & 1 & -3 \end{array} \right]$$

$$R_3 \longrightarrow 11R_3 + R_2$$

$$[A : B] \approx \left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The augmented matrix $[A : B]$ is in row echelon form, $\rho(A) = 2$ and $\rho(A : B) = 2$.

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

$\therefore \rho(A : B) = \rho(A) = 2$ and number of unknowns in the given system of a linear equations are $n = 3$.

Therefore, the given system of a linear equations are consistent and has an infinitely many solutions. Let us assume that $n - r = 3 - 2 = 1$ unknown quantities are an arbitrary constant k . Let $z = k$

$$5x + 3y + 7z = 4 \quad (4)$$

$$121y - 11z = 33 \quad (5)$$

Now, solving equations (4) and (5)

From Eqn. (5)

$$121y = 33 + 11k$$

dividing by 11

$$11y = 3 + k$$

$$\rightarrow y = \frac{3}{11} + \frac{k}{11}$$

$$\text{From Eqn, (4)} \quad 5x + 3\left(\frac{3}{11} + \frac{k}{11}\right) + 7k = 4$$

$$\rightarrow x = \frac{35-80k}{55}$$

$\therefore x = \frac{35-80k}{55}$, $y = \frac{3+k}{11}$ and $z = k$ are the infinitely many solutions for the given system of a linear equations.

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

Problem - 03: Show that the system of a linear equations $x + y + z = -3$, $3x + y - 2z = -2$, $2x + 4y + 7z = 7$ is not consistent.

Solution: Given the system of linear equations

$$x + y + z = -3, 3x + y - 2z = -2, 2x + 4y + 7z = 7$$

The given system of a linear equations are in matrix equation

$$AX = B$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & -2 \\ 2 & 4 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ -2 \\ 7 \end{bmatrix}$$

The augmented matrix of the given system of linear equations $[A : B]$ is

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 3 & 1 & -2 & : & -2 \\ 2 & 4 & 7 & : & 7 \end{bmatrix}$$

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

Reducing the augmented matrix $[A : B]$ to row echelon form by applying the elementary row transformations.

$$R_2 \longrightarrow R_2 - 3R_1$$

$$R_3 \longrightarrow R_3 - 2R_1$$

$$[A : B] \approx \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 2 & 5 & : & 13 \end{bmatrix}$$

$$R_3 \longrightarrow R_3 + R_2$$

$$[A : B] \approx \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 0 & 0 & : & 20 \end{bmatrix}$$

The augmented matrix $[A : B]$ is in row echelon form, $\rho(A) = 2$ and $\rho(A : B) = 3$.
 $\therefore \rho(A) \neq \rho(A : B)$, the given system of a linear equations has no solution and hence it not consistent.

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

Problem - 04: Find the value of λ for which the system of a linear equations $x + y + z = 1$, $x + 2y + 4z = \lambda$, $x + 4y + 10z = \lambda^2$ has a solution and solve it in each case.

Solution: Given the system of a linear equations

$$x + y + z = 1, x + 2y + 4z = \lambda, x + 4y + 10z = \lambda^2$$

The given system of a linear equations are in matrix equation

$$AX = B$$
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \lambda \\ \lambda^2 \end{bmatrix}$$

The augmented matrix of the given system of linear equations $[A : B]$ is

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & \lambda \\ 1 & 4 & 10 & \lambda^2 \end{array} \right]$$

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

Reducing the augmented matrix $[A : B]$ to row echelon form by applying the elementary row transformations.

$$R_2 \longrightarrow R_2 - R_1$$

$$R_3 \longrightarrow R_3 - R_1$$

$$[A : B] \approx \left[\begin{array}{cccc|c} 1 & 1 & 1 & & 1 \\ 0 & 1 & 3 & & \lambda - 1 \\ 0 & 3 & 9 & & \lambda^2 - 1 \end{array} \right]$$

$$R_3 \longrightarrow R_3 - 3R_2$$

$$[A : B] \approx \left[\begin{array}{cccc|c} 1 & 1 & 1 & & 1 \\ 0 & 1 & 3 & & \lambda - 1 \\ 0 & 0 & 0 & & \lambda^2 - 3\lambda + 2 \end{array} \right]$$

The augmented matrix $[A : B]$ is in row echelon form, $\rho(A) = 2$ and $\rho(A : B) = 3$, but the given system of a linear equations has a solution if $\rho(A : B) = 2$ only when $\lambda^2 - 3\lambda + 2 = 0$. Thus, the given system of a linear equations is consistent for $\lambda = 1$ or $\lambda = 2$

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

Case - 01: When $\lambda = 1$ and the given system of a linear equations reduces to

$$x + y + z = 1$$

$$y + 3z = 0$$

From these equations we get, $z = k_1$, $y = -3k_1$ and $x = 1 - 2k_1$ are the infinitely many solutions for the given system of a linear equations. Where k_1 be a arbitrary constant.

Case - 02: When $\lambda = 2$ and the given system of a linear equations reduces to

$$x + y + z = 1$$

$$y + 3z = 1$$

From these equations we get, $z = k_2$, $y = 1 - 3k_2$ and $x = 2k_2$ are the infinitely many solutions for the given system of a linear equations. Where k_2 be a arbitrary constant.

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

Problem - 05: Find the values of λ and μ for which the system of a linear equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ has (i) a unique solution, (ii) an infinitely solution and (iii) no solution.

Solution: Given the system of a linear equations

$$x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu$$

The given system of a linear equations are in matrix equation

$$AX = B$$
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$$

The augmented matrix of the given system of linear equations $[A : B]$ is

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix}$$

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

Reducing the augmented matrix $[A : B]$ to row echelon form by applying the elementary row transformations.

$$R_2 \longrightarrow R_2 - R_1$$

$$R_3 \longrightarrow R_3 - R_1$$

$$[A : B] \approx \left[\begin{array}{cccc|c} 1 & 1 & 1 & & 6 \\ 0 & 1 & 2 & & 4 \\ 0 & 1 & \lambda - 1 & & \mu - 6 \end{array} \right]$$

$$R_3 \longrightarrow R_3 - R_2$$

$$[A : B] \approx \left[\begin{array}{cccc|c} 1 & 1 & 1 & & 6 \\ 0 & 1 & 2 & & 4 \\ 0 & 0 & \lambda - 3 & & \mu - 10 \end{array} \right]$$

From this, we observed that the rank of the coefficient matrix A depends on the value of λ and that the rank of augmented matrix $[A : B]$ depends on the values of both λ and μ .

Continued Unit - IV Problems on Consistent of a system of Linear Equations ...

- (i) **An unique solution:** The given system of a linear equations is consistent and has an unique solution if $\rho(A) = \rho(A : B) = r = n$. This is possible only when $\lambda \neq 3$, any value of μ .
- (ii) **An infinitely many solutions:** The given system of a linear equations is consistent and has an infinitely many solutions if $\rho(A) = \rho(A : B) = r < n$. This is possible only when $\lambda = 3$ and $\mu = 10$.
- (iii) **No solution:** The given system of a linear equations is inconsistent or no solution if $\rho(A) \neq \rho(A : B)$. This is possible only when $\lambda = 3$ and $\mu \neq 10$.

Continued Unit - IV on Gauss - Elimination Method to solve a system of a linear equations ...

Gauss - Elimination Method:

Gauss - Elimination method is a basic numerical method to solve system of a linear equations. By using Gauss - elimination method, we have to reducing the augmented matrix $[A : B]$ to an upper triangular matrix $[U : C]$, by applying elementary row transformations. Where U is the upper triangular matrix and C is the column matrix with constants. Solve the unknown quantities (or unknown Variables) using back substitution method.

Problem - 01: Solve the system of a linear equations by Gauss - elimination method

$$2x_1 + x_2 + 4x_3 = 12$$

$$4x_1 + 11x_2 - x_3 = 33$$

$$8x_1 - 3x_2 + 20x_3 = 20$$

Solution: The augmented matrix for the given system of linear equations is

$$[A : B] = \left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 4 & 11 & -1 & 33 \\ 8 & -3 & 20 & 20 \end{array} \right]$$

Reducing the augmented matrix $[A : B]$ to row echelon form by applying the elementary row transformations.

Continued Unit - IV Problems on Gauss - Elimination Method ...

$$[A : B] \approx \left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 0 & 9 & -9 & 9 \\ 0 & -7 & 4 & -28 \end{array} \right]$$

$$R_2 \longrightarrow \frac{1}{9}R_2$$

$$[A : B] \approx \left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 0 & 1 & -1 & 1 \\ 0 & -7 & 4 & -28 \end{array} \right]$$

$$R_3 \longrightarrow R_3 + 7R_2$$

$$[A : B] \approx \left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & -3 & -21 \end{array} \right]$$

Now, converted to a system of linear equations

$$2x_1 + x_2 + 4x_3 = 12 \quad (6)$$

$$x_2 - x_3 = 1 \quad (7)$$

$$-3x_3 = -21 \quad (8)$$

Continued Unit - IV Problems on Gauss - Elimination Method ...

solve by back substitution method

From Eqn. (8), $x_3 = 7$, Eqn. (7) $x_2 - 7 = 1$, $x_2 = 8$ and Eqn. (6)

$$2x_1 + 8 + 4(7) = 12, x_1 = -12$$

$\therefore x_1 = -12$, $x_2 = 8$ and $x_3 = 7$ are the solutions for the given system of a linear equations.

Problem - 02: Solve the system of a linear equations by Gauss - elimination method

$$2x + 5y + 7z = 52$$

$$2x + y - z = 0$$

$$x + y + z = 9$$

Solution: The augmented matrix for the given system of linear equations is

$$[A : B] = \left[\begin{array}{ccc|c} 2 & 5 & 7 & 52 \\ 2 & 1 & -1 & 0 \\ 1 & 1 & 1 & 9 \end{array} \right]$$

Reducing the augmented matrix $[A : B]$ to row echelon form by applying the elementary row transformations.

$$R_2 \longrightarrow R_2 - R_1$$

$$R_3 \longrightarrow 2R_3 - R_1$$

Continued Unit - IV Problems on Gauss - Elimination Method ...

$$[A : B] \approx \left[\begin{array}{ccc|c} 2 & 5 & 7 & 52 \\ 0 & -4 & -8 & -52 \\ 0 & -3 & -5 & -43 \end{array} \right]$$

$$R_2 \longrightarrow \frac{-1}{4}R_2$$

$$[A : B] \approx \left[\begin{array}{ccc|c} 2 & 5 & 7 & 52 \\ 0 & 1 & 2 & 13 \\ 0 & -3 & -5 & -43 \end{array} \right]$$

$$R_3 \longrightarrow R_3 + 3R_2$$

$$[A : B] \approx \left[\begin{array}{ccc|c} 2 & 5 & 7 & 52 \\ 0 & 1 & 2 & 13 \\ 0 & 0 & 1 & -4 \end{array} \right]$$

Now, converted to a system of linear equations

$$2x + 5y + 7z = 52 \quad (9)$$

$$y + z = 13 \quad (10)$$

$$z = -4 \quad (11)$$

solve by back substitution method

From Eqn. (11), $z = -4$, Eqn. (10) $y + 4 = 13$, $y = 9$ and Eqn. (9) $x + 9 - 4 = 52$,
 $x = 47$

$\therefore x = 47$, $y = 9$ and $z = -4$ are the solutions for the given system of a linear equations.

Problems: Solve the following system of linear equations using Gauss - elimination method.

(1)

$$2x + 5y + 7z = 52, 2x + y - z = 0, x + y + z = 9$$

(2)

$$2x_1 - x_2 + 3x_3 = 1, -3x_1 + 4x_2 - 5x_3 = 0, x_1 + 3x_2 - 6x_3 = 0$$

(3)

$$x + 2y - z = 3, 3x - y + 2z = 1, 2x - 2y + 3z = 2$$

Continued Unit - IV on Gauss - Jacobi's iterative Method to solve a system of a linear equations ...

Diagonally dominant system of a linear equations: A system of a linear equations is said to be in diagonally dominant system in which the numerically large coefficients are along the leading diagonal of the coefficient matrix. This kind of dominant system of a linear equations are solve by the following two methods, they are,

- 1 Gauss - Jacobi's iterative method.
- 2 Gauss - Seidel iterative method.

Consider the system of equations as follows

$$a_{11}x + a_{12}y + a_{13}z = b_1 \quad (12)$$

$$a_{21}x + a_{22}y + a_{23}z = b_2 \quad (13)$$

$$a_{31}x + a_{32}y + a_{33}z = b_3 \quad (14)$$

The conditions to verify the above system of a linear equations are in diagonally dominant system, if it satisfies the following conditions,

$$|a_{11}| > |a_{12}| + |a_{13}|$$

$$|a_{22}| > |a_{21}| + |a_{23}|$$

$$|a_{33}| > |a_{31}| + |a_{32}|$$

Continued Unit - IV on Gauss - Jacobi's iterative Method to solve a system of a linear equations ...

Note - 01: First verify the given system of a linear equations are in diagonally dominant form. Suppose the given system of a linear equations are not in diagonally dominant form then rearrange the given system of a linear equations to diagonally dominant form and by using Gauss - Jacobi's method OR Gauss - Seidel method.

Note - 02: Suppose if an initial approximation is not given then by taking an initial approximations all are $(0, 0, 0)$.

I - Gauss - Jacobi's iterative Method:

This is a numerically an iterative method to obtain an approximate solution of the given system of a linear equations and it is applicable only a diagonally dominant system of a linear equations.

Continued Unit - IV problems on Gauss - Jacobi's iterative Method to solve a system of a linear equations ...

Problem - 01: Use Gauss - Seidel iteration method to solve the system of a linear equations.

$$10x + y + z = 12, 2x + 10y + z = 13, 2x + 2y + 10z = 14$$

Solution: By data the system of a linear equations are

$$10x + y + z = 12$$

$$2x + 10y + z = 13$$

$$2x + 2y + 10z = 14$$

We can observe that the given system of a linear equations are in diagonally dominant form and we can solve x , y and z as follows

$$x^{(n+1)} = \frac{1}{10}[12 - y^{(n)} - z^{(n)}] \quad (15)$$

$$y^{(n+1)} = \frac{1}{10}[13 - 2x^{(n)} - z^{(n)}] \quad (16)$$

$$z^{(n+1)} = \frac{1}{10}[14 - 2x^{(n)} - 2y^{(n)}] \quad (17)$$

Continued Unit - IV problems on Gauss - Jacobi's iterative Method to solve a system of a linear equations ...

Let us choose an initial approximations are $x^0 = 0$, $y^0 = 0$ and $z^0 = 0$.

I - iteration:

Substitute $y = y^{(0)} = 0$ and $z = z^{(0)} = 0$ in Eqn. (15)

$$x^{(1)} = \frac{1}{10}(12 - 0 - 0) = \frac{12}{10} = 1.2$$

Substitute $x = x^{(1)} = 1.2$ and $z = z^{(0)} = 0$ in Eqn. (16)

$$y^{(1)} = \frac{1}{10}(13 - 2(1.2) - 0) = \frac{10.6}{10} = 1.06$$

Substitute $x = x^{(1)} = 1.2$ and $y = y^{(1)} = 1.06$ in Eqn. (17)

$$z^{(1)} = \frac{1}{10}(14 - 2(1.2) - 2(1.06)) = 0.948$$

II - iteration:

Substitute $y = y^{(1)} = 1.06$ and $z = z^{(1)} = 0.948$ in Eqn. (15)

$$x^{(2)} = \frac{1}{10}(12 - 1.06 - 0.948) = \frac{9.992}{10} = 0.9992$$

Continued Unit - IV problems on Gauss - Jacobi's iterative Method to solve a system of a linear equations ...

Continued Unit - IV problems on Gauss - Jacobi's iterative Method to solve a system of a linear equations ...

Continued Unit - IV on Gauss - Seidel Iterative Method to solve a system of a linear equations ...

II - Gauss - Seidel Iterative Method:

The Gauss - Seidel method is also a numerically an iterative method to find an approximate solution of the given system of a linear equations. This method is applicable for only the diagonally dominant system of a linear equations.

Continued Unit - IV Problems on Gauss - Seidel Iterative Method ...

Problem - 01: Use Gauss - Seidel iteration method to solve the system of a linear equations.

$$10x + y + z = 12, 2x + 10y + z = 13, 2x + 2y + 10z = 14$$

Solution: By data the system of a linear equations are

$$10x + y + z = 12$$

$$2x + 10y + z = 13$$

$$2x + 2y + 10z = 14$$

We can observe that the given system of a linear equations are in diagonally dominant form and solve x , y and z as follows

$$x^{(n+1)} = \frac{1}{10}[12 - y^{(n)} - z^{(n)}] \quad (18)$$

$$y^{(n+1)} = \frac{1}{10}(13 - 2x^{(n+1)} - z^{(n)}) \quad (19)$$

$$z^{(n+1)} = \frac{1}{10}(14 - 2x^{(n+1)} - 2y^{(n+1)}) \quad (20)$$

Continued Unit - IV Problems on Gauss - Seidel Iterative Method ...

An initial approximation is not given in the problem and let us assume that an initial approximations are $x^0 = 0$, $y^0 = 0$ and $z^0 = 0$.

I - iteration: Substitute $n = 0$

Substitute $y = y^{(0)} = 0$ and $z = z^{(0)} = 0$ in Eqn. (15)

$$x^{(1)} = \frac{1}{10}(12 - 0 - 0) = \frac{12}{10} = 1.2$$

Substitute $x = x^{(1)} = 1.2$ and $z = z^{(0)} = 0$ in Eqn. (16)

$$y^{(1)} = \frac{1}{10}(13 - 2(1.2) - 0) = \frac{10.6}{10} = 1.06$$

Substitute $x = x^{(1)} = 1.2$ and $y = y^{(1)} = 1.06$ in Eqn. (17)

$$z^{(1)} = \frac{1}{10}(14 - 2(1.2) - 2(1.06)) = 0.948$$

II - iteration:

Substitute $y = y^{(1)} = 1.06$ and $z = z^{(1)} = 0.948$ in Eqn. (15)

$$x^{(2)} = \frac{1}{10}(12 - 1.06 - 0.948) = \frac{9.992}{10} = 0.9992$$

Continued Unit - IV Problems on Gauss - Seidel Iterative Method ...

Substitute $x = x^{(2)} = 0.9992$ and $z = z^{(1)} = 0.948$ in Eqn. (16)

$$y^{(2)} = \frac{1}{10}(13 - 2(0.9992) - 0.948) = \frac{10.0536}{10} = 1.00536$$

Substitute $x = x^{(2)} = 0.9992$ and $y = y^{(2)} = 1.0536$ in Eqn. (17)

$$z^{(2)} = \frac{1}{10}(14 - 2(0.9992) - 2(1.0536)) = 0.98944$$

III - iteration:

Substitute $y = y^{(2)} = 1.00536$ and $z = z^{(2)} = 0.98944$ in Eqn. (15)

$$x^{(3)} = \frac{1}{10}(12 - 1.00536 - 0.98944) = 1.00052$$

Substitute $x = x^{(3)} = 1.00052$ and $z = z^{(2)} = 0.98944$ in Eqn. (16)

$$y^{(3)} = \frac{1}{10}(13 - 2(1.00052) - 0.98944) = 1.000952$$

Substitute $x = x^{(3)} = 1.00052$ and $y = y^{(3)} = 1.000952$ in Eqn. (17)

$$z^{(3)} = \frac{1}{10}(14 - 2(1.00052) - 2(1.000952)) = 0.9997$$

IV - iteration:

Substitute $y = y^{(3)} = 1.000952$ and $z = z^{(3)} = 0.9997$ in Eqn. (15)

$$x^{(4)} = \frac{1}{10}(12 - 1.000952 - 0.9997) = 0.999934$$

Substitute $x = x^{(4)} = 0.999934$ and $z = z^{(3)} = 0.9997$ in Eqn. (16)

$$y^{(4)} = \frac{1}{10}(13 - 2(0.999934) - 0.9997) = 1.0000432$$

Substitute $x = x^{(4)} = 0.999934$ and $y = y^{(4)} = 1.000043$ in Eqn. (17)

$$z^{(4)} = \frac{1}{10}(14 - 2(0.999934) - 2(1.000043)) = 1.000000456$$

Hence, $x = 0.999934 \approx 1$, $y = 1.0000432 \approx 1$ and $z = 1.000000456 \approx 1$ are the solutions for the given system of a linear equations.

Continued Unit - IV Problems on Gauss - Seidel Method ...

Problem - 02: Apply Gauss - Seidel iteration method to solve the system of a linear equations.

$$-x - y - 2z + 10u = -9, -2x + 10y - z - u = 15, -x - y + 10z - 2u = 27, 10x - 2y - z - u = 3$$

Solution: By data the system of a linear equations are

$$-x - y - 2z + 10u = -9$$

$$-2x + 10y - z - u = 15$$

$$-x - y + 10z - 2u = 27$$

$$10x - 2y - z - u = 3$$

We can observe that the given system of a linear equations is not in diagonally dominant form and we can rearrange it to reach the diagonally dominant form, that is

$$10x - 2y - z - u = 3$$

$$-2x + 10y - z - u = 15$$

$$-x - y + 10z - 2u = 27$$

$$-x - y - 2z + 10u = -9$$

Continued Unit - IV Problems on Gauss - Seidel Method ...

Now, from these system of a linear equations, we can solve x , y , z and u , we get

$$x = \frac{1}{10}(3 + 2y + z + u) \quad (21)$$

$$y = \frac{1}{10}(15 + 2x + z + u) \quad (22)$$

$$z = \frac{1}{10}(27 + x + y + 2u) \quad (23)$$

$$u = \frac{1}{10}(-9 + x + y + 2z) \quad (24)$$

Taking the initial approximations are $x^{(0)} = 0$, $y^{(0)} = 0$, $z^{(0)} = 0$ and $u^{(0)} = 0$.

Iteration - I:

Substitute $y = y^{(0)} = 0$, $z = z^{(0)} = 0$ and $u = u^{(0)} = 0$ in Eqn. (18)

$$x^{(1)} = \frac{1}{10}(3 + 2(0) + (0) + (0)) = 0.3$$

Substitute $x = x^{(1)} = 0.3$, $z = z^{(0)} = 0$ and $u = u^{(0)} = 0$ in Eqn. (19)

$$y^{(1)} = \frac{1}{10}(15 + 2(0.3) + (0) + (0)) = 1.56$$

Continued Unit - IV Problems on Gauss - Seidel Method ...

Substitute $x = x^{(1)} = 0.3$, $y = y^{(1)} = 1.56$ and $u = u^{(0)} = 0$ in Eqn. (20)

$$z^{(1)} = \frac{1}{10}(27 + (0.3) + (1.56) + 2(0)) = 2.886$$

Substitute $x = x^{(1)} = 0.3$, $y = y^{(1)} = 1.56$ and $z = z^{(1)} = 2.886$ in Eqn. (21)

$$u^{(1)} = \frac{1}{10}(-9 + (0.3) + (1.56) + 2(2.884)) = -0.1368$$

Iteration - II:

Substitute $y = y^{(1)} = 1.56$, $z = z^{(1)} = 2.886$ and $u = u^{(1)} = -0.1386$ in Eqn. (18)

$$x^{(2)} = \frac{1}{10}(3 + 2(1.56) + (2.886) + (-0.1368)) = 0.88692$$

Substitute $x = x^{(2)} = 0.88692$, $z = z^{(1)} = 2.886$ and $u = u^{(1)} = -0.1368$ in Eqn. (19)

$$y^{(2)} = \frac{1}{10}(15 + 2(0.88692) + (2.886) + (-0.1368)) = 1.952304$$

Continued Unit - IV Problems on Gauss - Seidel Method ...

Substitute $x = x^{(2)} = 0.88692$, $y = y^{(2)} = 1.952304$ and $u = u^{(1)} = -0.1368$ in Eqn. (20)

$$z^{(2)} = \frac{1}{10}(27 + (0.88692) + (1.952304) + 2(-0.1368)) = 2.95656$$

Substitute $x = x^{(2)} = 0.88692$, $y = y^{(2)} = 1.952304$ and $z = z^{(2)} = 2.95656$ in Eqn. (21)

$$u^{(2)} = \frac{1}{10}(-9 + (0.88692) + (1.952304) + 2(2.95656)) = -0.02476$$

Iteration - III:

Substitute $y = y^{(2)} = 1.952304$, $z = z^{(2)} = 2.95656$ and $u = u^{(2)} = -0.02476$ in Eqn. (18)

$$x^{(3)} = \frac{1}{10}(3 + 2(1.952304) + (2.95656) + (-0.02476)) = 0.9836$$

Continued Unit - IV Problems on Gauss - Seidel Method ...

Substitute $x = x^{(3)} = 0.9836$, $z = z^{(2)} = 2.95656$ and $u = u^{(2)} = -0.02476$ in Eqn. (19)

$$y^{(3)} = \frac{1}{10}(15 + 2(0.9836) + (2.95656) + (-0.02476)) = 1.9899$$

Substitute $x = x^{(3)} = 0.9836$, $y = y^{(3)} = 1.9899$ and $u = u^{(2)} = -0.02476$ in Eqn. (20)

$$z^{(2)} = \frac{1}{10}(27 + (0.9836) + (1.9899) + 2(-0.02476)) = 2.99239$$

Substitute $x = x^{(3)} = 0.9836$, $y = y^{(3)} = 1.9899$ and $z = z^{(3)} = 2.99239$ in Eqn. (21)

$$u^{(2)} = \frac{1}{10}(-9 + (0.9836) + (1.9899) + 2(2.99239)) = -0.004172$$

Proceeding in this way, we get

Iteration - IV:

$$x^{(4)} = 0.9968, y^{(4)} = 1.9982, z^{(4)} = 2.9987, u^{(4)} = -0.00038$$

Continued Unit - IV Problems on Gauss - Seidel Method ...

Iteration - V:

$$x^{(5)} = 0.9994, y^{(5)} = 1.9997, z^{(5)} = 2.9998, u^{(5)} = -0.0001$$

Iteration - VI:

$$x^{(6)} = 0.9999, y^{(6)} = 1.9999, z^{(6)} = 3.0000, u^{(6)} = 0.0000$$

Hence, $x = 1$, $y = 2$, $z = 3$ and $u = 0$ are the solutions of the given system of a linear equations.

Problem - 03: Apply Gauss - Seidel iteration method to solve the system of a linear equations

$$x + y + 54z = 110, 27x + 6y - z = 85, 6x + 15y + 2z = 72$$

Perform three iterations.

Solution: By data the system of a linear equations are

$$x + y + 54z = 110$$

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

Continued Unit - IV Problems on Gauss - Seidel Method ...

We can observe that the given system of a linear equations are not in diagonally dominant. Therefore, rearrange the given system of a linear equations as

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

Now, solve x , y and z we get

$$x = \frac{1}{27}(85 - 6y + z) \quad (25)$$

$$y = \frac{1}{15}(72 - 6x - 2z) \quad (26)$$

$$z = \frac{1}{54}(110 - x - y) \quad (27)$$

By taking an initial approximations are $x^0 = 0$, $y^0 = 0$ and $z^0 = 0$.

I - iteration:

Substitute $y = y^{(0)} = 0$ and $z = z^{(0)} = 0$ in Eqn. (22)

$$x^{(1)} = \frac{1}{27}(85 - 0 - 0) = \frac{85}{27} = 3.1481$$

Continued Unit - IV Problems on Gauss - Seidel Method ...

Substitute $x = x^{(1)} = 3.1481$ and $z = z^{(0)} = 0$ in Eqn. (23)

$$y^{(1)} = \frac{1}{15}(72 - 6(3.1481) - 0) = 3.54076$$

Substitute $x = x^{(1)} = 3.1481$ and $y = y^{(1)} = 3.54076$ in Eqn. (24)

$$z^{(1)} = \frac{1}{54}(110 - 3.1481 - 3.54076) = 1.9131$$

II - iteration:

Substitute $y = y^{(1)} = 3.54076$ and $z = z^{(1)} = 1.9131$ in Eqn. (22)

$$x^{(2)} = \frac{1}{27}(85 - (6 \times 3.54076) + 1.9131) = 2.4321$$

Substitute $x = x^{(2)} = 2.4321$ and $z = z^{(1)} = 1.9131$ in Eqn. (23)

$$y^{(2)} = \frac{1}{15}(72 - 6(2.4321) - 1.9131) = 3.69962$$

Substitute $x = x^{(2)} = 2.4321$ and $y = y^{(2)} = 3.69962$ in Eqn. (24)

$$z^{(2)} = \frac{1}{54}(110 - 2.4321 - 3.69962) = 1.9234$$

III - iteration:

Substitute $y = y^{(2)} = 3.69962$ and $z = z^{(2)} = 1.9234$ in Eqn. (22)

$$x^{(3)} = \frac{1}{27}(85 - (6 \times 3.69962) + 1.9234) = 2.3972$$

Substitute $x = x^{(3)} = 2.3972$ and $z = z^{(2)} = 1.9234$ in Eqn. (23)

$$y^{(3)} = \frac{1}{15}(72 - 6(2.3972) - 1.9234) = 3.7128$$

Substitute $x = x^{(3)} = 2.3972$ and $y = y^{(3)} = 3.7128$ in Eqn. (24)

$$z^{(3)} = \frac{1}{54}(110 - 2.3972 - 3.7128) = 1.9238$$

Hence, $x = 2.3972$, $y = 3.7128$, $z = 1.9238$ are the solutions of the given system of a linear equations.

Continued Unit - IV Problems on Gauss - Seidel Method ...

Problems: Apply Gauss - Seidel iteration method to solve the following systems of a linear equations

(1)

$$20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25$$

(2)

$$10x + 2y + z = 9, 2x + 20y - 2z = -44, -2x + 3y + 10z = 22$$

(3)

$$5x + 2y + z = 12, x + 4y + 2z = 15, x + 2y + 5z = 20$$

(4)

$$13x + 5y - 3z + u = 18, 2x + 12y + z - 4u = 13, 3x - 4y + 10z + u = 29, 2x + y - 3z + 9u = 31$$

Continued Uni - IV on Eigen values and Eigen vector of a Matrix ...

Definition of a Eigen value and Eigen vector of the matrix:

Let us take a square matrix A of order $m \times n$, if there exists a scalar constant λ and a non zero column matrix X such that $AX = \lambda X$, then λ is known as an eigenvalue of a matrix A and X is known as an eigenvector of a matrix A corresponding to an eigenvalue λ .

Example: Consider a square matrix

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

and

$$X = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$$

then

$$AX = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ -6 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} = \lambda X$$

Continued Unit - IV on Eigen values and Eigen vector of a Matrix ...

Therefore, $\lambda = 3$ is called an eigen value of a matrix A and $X = [1, -1, -2]^T$ is called an eigen vector of a matrix A corresponding to an eigen value $\lambda = 3$.

Let I be the unit matrix and $X = IX$, then we can write $AX = \lambda X$ as,

$$AX = \lambda IX$$

$$\rightarrow AX - \lambda IX = 0$$

$$\rightarrow (A - \lambda I)X = 0$$

Where,

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

and

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

and

$$\lambda I = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

continued Unit - IV on Eigen values and Eigen vector of a Matrix ...

Therefore, $(A - \lambda I)X = 0$ becomes in the matrix form

$$[A - \lambda I]X = \begin{bmatrix} a_1 - \lambda & b_1 & c_1 \\ a_2 & b_2 - \lambda & c_2 \\ a_3 & b_3 & c_3 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(a_1 - \lambda)x_1 + b_1x_2 + c_1x_3 = 0$$

$$a_2x_1 + (b_2 - \lambda)x_2 + c_2x_3 = 0$$

$$a_3x_1 + b_3x_2 + (c_3 - \lambda)x_3 = 0$$

Therefore, this homogeneous system of linear equations has a non trivial solution exists if the determinant of the co efficient matrix is equal to zero.

$$\begin{vmatrix} a_1 - \lambda & b_1 & c_1 \\ a_2 & b_2 - \lambda & c_2 \\ a_3 & b_3 & c_3 - \lambda \end{vmatrix} = 0$$

$\therefore |A - \lambda I| = 0$ is called as a characteristics equation.

On expanding this determinant, we get a polynomial equation in λ of degree 3 is called characteristics equation. The roots of this characteristics equation are called as characteristics roots or Eigen roots or Eigen values of the matrix A . For each eigenvalue λ , there will be an eigenvector $X \neq 0$, is called characteristic vector or eigenvector of the matrix A .

Continued Unit - II on Properties of an Eigenvalues and Eigenvector of a Matrix ...

Property - 01: If λ be an eigenvalue of a square matrix A with X as a corresponding eigenvector, then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} .

Proof: We know that, by definition of an eigenvalue and eigenvector of a square matrix A is

$$AX = \lambda X \quad (28)$$

Pre - multiplying by A^{-1} on both sides, we get

$$A^{-1}(AX) = A^{-1}(\lambda X)$$

$$(A^{-1}A)X = (\lambda)(A^{-1}X)$$

But, $A^{-1}A = I$ and $IX = X$

$$(I)X = (\lambda)(A^{-1}X)$$

$$IX = (\lambda)(A^{-1}X)$$

$$X = (\lambda)(A^{-1}X)$$

$$A^{-1}X = \left(\frac{1}{\lambda}\right)X \quad (29)$$

Continued Unit - IV on Properties of an Eigenvalues and Eigenvector of a Matrix ...

Comparing the Eqns. (14) and (15) and hence we can conclude that, $\frac{1}{\lambda}$ is a eigenvalue of a A^{-1} .

Property - 02: If λ be an eigenvalue of a square matrix A with X as a corresponding eigenvector, then λ^2 is an eigenvalue of A^2 and hence λ^m is an eigenvalue of A^m , where m is a positive integer.

Proof: We know that, by definition of an eigenvalue and eigenvector of a square matrix A is

$$AX = \lambda X \quad (30)$$

Multiplying by A on both sides, we get

$$A(AX) = A(\lambda X)$$

$$(A^2)X = (\lambda)(AX)$$

But $AX = \lambda X$

$$(A^2)X = (\lambda)(\lambda X)$$

$$A^2X = \lambda^2X \quad (31)$$

Comparing the Eqns. (16) and (17) and hence we can conclude that, λ^2 is a eigenvalue of a A^2 .

Continued Unit - IV on Properties of an Eigenvalues and Eigenvector of a Matrix ...

Again, multiplying by A on both sides, we get

$$A(A^2X) = (A)\lambda^2X$$

$$A^3X = \lambda^2(AX)$$

But $AX = \lambda X$

$$A^3X = \lambda^2(\lambda X)$$

$$A^3X = \lambda^3X$$

Therefore, λ^3 is a eigenvalue of a A^3 . Similarly, continue like this m times, we get

$$A^mX = \lambda^mX$$

Hence, λ^m is a eigenvalue of a A^m .

Property - 03: A matrix A and its transpose A^T have same eigenvalues.

Continued Unit - IV on Properties of an Eigenvalues and Eigenvector of a Matrix ...

Property - 03: If A is an upper triangular matrix with X as a corresponding eigen vector then the eigenvalue of A are the just the diagonal elements of the matrix A .

Proof: Consider an upper triangular matrix A of the form

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ 0 & b_2 & c_2 \\ 0 & 0 & c_3 \end{bmatrix}$$

The characteristics equation of a matrix A is

$$|A - \lambda X| = 0$$

$$\begin{vmatrix} (a_1 - \lambda) & b_1 & c_1 \\ 0 & (b_2 - \lambda) & c_2 \\ 0 & 0 & (c_3 - \lambda) \end{vmatrix} = 0$$

Now, expanding the above determinant we get

$$(a_1 - \lambda)[(b_2 - \lambda)(c_3 - \lambda) - 0] - b_1[0 - 0] + c_1[0 - 0] = 0$$

$$(a_1 - \lambda)(b_2 - \lambda)(c_3 - \lambda) = 0$$

Continued Unit - IV on Properties of an Eigenvalues and Eigenvector of a Matrix ...

Therefore,

$$\lambda = a_1, \lambda = b_2, \lambda = c_3$$

a_1 , b_2 and c_3 are the diagonal elements of the matrix A and it is a triangular matrix. Hence the eigenvalues of the triangular matrix are just the diagonal elements of the matrix.

Property - 04: The sum of the eigenvalues of the matrix A is equal to the trace of the matrix A .

Note: The trace of the matrix A is equal to the sum of the diagonal elements of the matrix A . $\therefore \text{Trace}(A) = \text{Sum of the diagonal elements of the matrix } A$.

Property - 05: The product of the eigenvalues of the matrix A is equal to the determinant of the matrix A .

$\therefore$ Product of the eigenvalues of the matrix $A = |A|$.

Working procedure to find an eigenvalues and eigenvectors of the given matrix:

Step - 01: Write the characteristic equation of the given matrix A

$$|A - \lambda I| = 0$$

Step - 02: Expanding the determinant, find the characteristics equation and roots of the characteristics equation are called an eigenvalues of the given matrix A .

Step - 03: Find the eigenvectors $X \neq 0$ for each eigenvalue λ of the given matrix.

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Problem - 01: Find all eigenvalues and corresponding eigenvectors of the matrix

$$\begin{bmatrix} -3 & 8 \\ -2 & 7 \end{bmatrix}$$

Solution: Take

$$A = \begin{bmatrix} -3 & 8 \\ -2 & 7 \end{bmatrix}$$

We know that, the characteristics equation of the given matrix is

$$\begin{aligned} |A - \lambda I| &= 0 \\ \begin{vmatrix} -3 - \lambda & 8 \\ -2 & 7 - \lambda \end{vmatrix} &= 0 \end{aligned}$$

Now, expanding this determinant, we get

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

$$(-3 - \lambda)(7 - \lambda) + 16 = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$(\lambda - 5)(\lambda + 1) = 0$$

$\therefore \lambda_1 = 5$ and $\lambda_2 = -1$ are the eigenvalues of the given matrix A .

Let $X = [x, y]^T$ be the eigenvector of the matrix A . The matrix equation is

$$[A - \lambda I][X] = [0]$$

$$\begin{bmatrix} -3 - \lambda & 8 \\ -2 & 7 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (32)$$

Case - 01: Put $\lambda = \lambda_1 = 5$ in Eqn. (18), we get

$$\begin{bmatrix} -8 & 8 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

From this we get the system of a linear equations

$$-8x + 8y = 0, -2x + 2y = 0,$$

The above equations are reduces to $x - y = 0$

$$x = y$$

$$\frac{x}{1} = \frac{y}{1} = k_1 (k_1 \neq 0)$$

where k_1 is a arbitrary constant, $x = k_1$ and $y = k_1$. Substitute x and y in the eigenvector X , we get the eigenvector $X_1 = [k_1, k_1]^T$ or $X_1 = [1, 1]$ if $k_1 = 1$ corresponding to the eigenvalue $\lambda = \lambda_1 = 5$.

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Case - 02: Put $\lambda = \lambda_2 = -1$ in Eqn. (18), we get

$$\begin{bmatrix} -2 & 8 \\ -2 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From this we get the system of a linear equations

$$-2x + 8y = 0, -2x + 8y = 0,$$

The above equations are reduces to $x - 4y = 0$

$$x = 4y$$

$$\frac{x}{4} = \frac{y}{1} = k_2 (k_2 \neq 0)$$

where k_2 is a arbitrary constant, $x = 4k_2$ and $y = k_2$. Substitute x and y in the eigenvector X , we get the eigenvector $X_2 = [4k_2, k_2]^T$ or $X_1 = [4, 1]$ if $k_2 = 1$ corresponding to the eigenvalue $\lambda = \lambda_2 = -1$.

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Problem - 02: Find all eigenvalues and corresponding eigenvectors of the matrix

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Solution: Take

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

We know that, the characteristics equation of the given matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2 - \lambda & 0 & 1 \\ 0 & 2 - \lambda & 0 \\ 1 & 0 & 2 - \lambda \end{vmatrix} = 0$$

Now, expanding this determinant, we get

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

$$(2 - \lambda)[(2 - \lambda)^2 - 0] - 0[0 - 0] + 1[0 - (2 - \lambda)] = 0$$

$$(2 - \lambda)[(2 - \lambda)^2] - (2 - \lambda) = 0$$

$$(2 - \lambda)[(2 - \lambda)^2 - 1] = 0$$

$$(2 - \lambda)(4 + \lambda^2 - 4\lambda - 1) = 0$$

$$(2 - \lambda)(\lambda^2 - 4\lambda + 3) = 0$$

$\rightarrow \lambda = 2, \lambda = 1$ and $\lambda = 3$

$\therefore \lambda = 1, \lambda = 2$ and $\lambda = 3$ are the eigen value of the given matrix A

Consider the system of matrix equation

$$[A - \lambda I][X] = [0]$$

where

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

$$\begin{bmatrix} 2-\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(2-\lambda)x_1 + 0x_2 + x_3 = 0$$

$$0x_1 + (2-\lambda)x_2 + 0x_3 = 0$$

$$1x_1 + 0x_2 + (2-\lambda)x_3 = 0$$

Case - 01: Substituting $\lambda = 1$ in the above system of a linear equations, we get

$$1x_1 + 0x_2 + x_3 = 0$$

$$0x_1 + 1x_2 + 0x_3 = 0$$

$$1x_1 + 0x_2 + 1x_3 = 0$$

solve by using cross multiplication method

$$\frac{x_1}{(0-1)} = -\frac{x_2}{(0-0)} = \frac{x_3}{(1-0)}$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Let k_1 be non zero arbitrary constant

$$\frac{x_1}{-1} = -\frac{x_2}{0} = \frac{x_3}{1} = k_1, (k_1 \neq 0)$$

$\therefore x_1 = -k_1, x_2 = 0$ and $x_3 = k_1$

Hence the eigen vector corresponding to eigen value $\lambda = 1$ is

$$X_1 = [-k_1, 0, k_1]^T = [-1, 0, 1]^T.$$

Case - 02: Substituting $\lambda = 2$ in the above system of a linear equations, we get

$$0x_1 + 0x_2 + x_3 = 0$$

$$0x_1 + 0x_2 + 0x_3 = 0$$

$$1x_1 + 0x_2 + 0x_3 = 0$$

From these, we get $x_1 = 0, x_3 = 0$ and let $x_2 = k_2$, where k_2 be a non zero arbitrary constant. Hence the eigenvector corresponding to the eigenvalue $\lambda = 2$ is

$$X_2 = [0, k_2, 0]^T = [0, 1, 0]^T.$$

Case - 03: Substituting $\lambda = 3$ in the above system of a linear equations, we get

$$(-1)x_1 + 0x_2 + x_3 = 0$$

$$0x_1 + (-1)x_2 + 0x_3 = 0$$

$$1x_1 + 0x_2 + (-1)x_3 = 0$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

From these, we get $-x_1 + x_3 = 0$, $-x_2 = 0$ and $x_1 - x_3 = 0$ let $x_1 = k_3$ and $x_3 = k_3$ where k_3 be a non zero arbitrary constant. Hence the eigenvector corresponding to the eigenvalue $\lambda = 3$ is $X_3 = [k_3, 0, k_3]^T = [1, 0, 1]^T$.

Problem - 02: Find all the Eigen values and Eigen vectors for the matrix

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Solution: Take

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

We know that, the characteristics equation of the given matrix is

$$|A - \lambda I| = 0$$
$$\begin{vmatrix} 8 - \lambda & -6 & 2 \\ -6 & 7 - \lambda & -4 \\ 2 & -4 & 3 - \lambda \end{vmatrix} = 0$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Problem - 02: Find all eigenvalues and corresponding eigenvectors of the matrix

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Solution: Take

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

We know that, the characteristics equation of the given matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2 - \lambda & 0 & 1 \\ 0 & 2 - \lambda & 0 \\ 1 & 0 & 2 - \lambda \end{vmatrix} = 0$$

Now, expanding this determinant, we get

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

$$(2 - \lambda)[(2 - \lambda)^2 - 0] - 0[0 - 0] + 1[0 - (2 - \lambda)] = 0$$

$$(2 - \lambda)[(2 - \lambda)^2] - (2 - \lambda) = 0$$

$$(2 - \lambda)[(2 - \lambda)^2 - 1] = 0$$

$$(2 - \lambda)(4 + \lambda^2 - 4\lambda - 1) = 0$$

$$(2 - \lambda)(\lambda^2 - 4\lambda + 3) = 0$$

$\rightarrow \lambda = 2, \lambda = 1$ and $\lambda = 3$

$\therefore \lambda = 1, \lambda = 2$ and $\lambda = 3$ are the eigen value of the given matrix A

Consider the system of matrix equation

$$[A - \lambda I][X] = [0]$$

where

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

$$\begin{bmatrix} 2-\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(2-\lambda)x_1 + 0x_2 + x_3 = 0$$

$$0x_1 + (2-\lambda)x_2 + 0x_3 = 0$$

$$1x_1 + 0x_2 + (2-\lambda)x_3 = 0$$

Case - 01: Substituting $\lambda = 1$ in the above system of a linear equations, we get

$$1x_1 + 0x_2 + x_3 = 0$$

$$0x_1 + 1x_2 + 0x_3 = 0$$

$$1x_1 + 0x_2 + 1x_3 = 0$$

solve by using cross multiplication method

$$\frac{x_1}{(0-1)} = -\frac{x_2}{(0-0)} = \frac{x_3}{(1-0)}$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Let k_1 be non zero arbitrary constant

$$\frac{x_1}{-1} = -\frac{x_2}{0} = \frac{x_3}{1} = k_1, (k_1 \neq 0)$$

$\therefore x_1 = -k_1, x_2 = 0$ and $x_3 = k_1$

Hence the eigen vector corresponding to eigen value $\lambda = 1$ is $X_1 = [-k_1, 0, k_1]^T$, if $k_1 = 1$, then $X_1 = [-1, 0, 1]^T$.

Case - 02: Substituting $\lambda = 2$ in the above system of a linear equations, we get

$$0x_1 + 0x_2 + x_3 = 0$$

$$0x_1 + 0x_2 + 0x_3 = 0$$

$$1x_1 + 0x_2 + 0x_3 = 0$$

From these, we get $x_1 = 0, x_3 = 0$ and let $x_2 = k_2$, where k_2 be a non zero arbitrary constant. Hence the eigenvector corresponding to the eigenvalue $\lambda = 2$ is $X_2 = [0, k_2, 0]^T$, if $k_2 = 1$, then $X_2 = [0, 1, 0]^T$.

Case - 03: Substituting $\lambda = 3$ in the above system of a linear equations, we get

$$(-1)x_1 + 0x_2 + x_3 = 0$$

$$0x_1 + (-1)x_2 + 0x_3 = 0$$

$$1x_1 + 0x_2 + (-1)x_3 = 0$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

From these, we get $-x_1 + x_3 = 0$, $-x_2 = 0$ and $x_1 - x_3 = 0$ let $x_1 = k_3$ and $x_3 = k_3$ where k_3 be a non zero arbitrary constant. Hence the eigenvector corresponding to the eigenvalue $\lambda = 3$ is $X_3 = [k_3, 0, k_3]^T$, if $k_3 = 1$, then $X_3 = [1, 0, 1]^T$.

Problem - 03: Find all the Eigen values and Eigen vectors for the matrix

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Solution: Take

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

We know that, the characteristics equation of the given matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 8 - \lambda & -6 & 2 \\ -6 & 7 - \lambda & -4 \\ 2 & -4 & 3 - \lambda \end{vmatrix} = 0$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Now, expanding this determinant, we get

$$(8 - \lambda)[(7 - \lambda)(3 - \lambda) - 16] + 6[-6(3 - \lambda) + 8] + 2[24 - 2(7 - \lambda)] = 0$$

$$(8 - \lambda)[\lambda^2 - 10\lambda + 5] + 6[6\lambda - 10] + 2[10 + 2\lambda] = 0$$

$$-\lambda^3 + 18\lambda^2 - 85\lambda + 40 + 40\lambda - 40 = 0$$

$$-\lambda(\lambda^2 - 18\lambda + 45) = 0$$

$\rightarrow \lambda = 0, \lambda = 15$ and $\lambda = 3$

$\therefore \lambda = 0, \lambda = 3$ and $\lambda = 15$ are the eigen value of the given matrix A

Consider the system of matrix equation

$$[A - \lambda I][X] = [0]$$

where

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (33)$$

Case - 01: Substituting $\lambda = 0$ in the above matrix equation (1), we get

$$8x_1 - 6x_2 + 2x_3 = 0$$

$$-6x_1 + 7x_2 - 4x_3 = 0$$

$$2x_1 - 4x_2 + 3x_3 = 0$$

solve by using cross multiplication method

$$\frac{x_1}{(24 - 14)} = -\frac{x_2}{(-32 + 12)} = \frac{x_3}{(56 - 36)}$$

$$\frac{x_1}{10} = \frac{x_2}{20} = \frac{x_3}{20}$$

$$\frac{x_1}{1} = -\frac{x_2}{2} = \frac{x_3}{2} = k_1, (k_1 \neq 0)$$

$\therefore x_1 = k_1, x_2 = 2k_1$ and $x_3 = 2k_1$. Substitute x_1, x_2 and x_3 in the eigenvector X , $X_1 = [k_1, 2k_1, 2k_1]^T$, if $k_1 = 1$, then $X_1 = [1, 2, 2]^T$ is eigenvector corresponding to the eigenvalue $\lambda = 0$ of the given matrix A .

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Case - 02: Substituting $\lambda = 3$ in the above matrix equation (1), we get

$$5x_1 - 6x_2 + 2x_3 = 0$$

$$-6x_1 + 4x_2 - 4x_3 = 0$$

$$2x_1 - 4x_2 + 0x_3 = 0$$

solve by using cross multiplication method

$$\frac{x_1}{(24 - 8)} = -\frac{x_2}{(-20 + 12)} = \frac{x_3}{(20 - 36)}$$

$$\frac{x_1}{16} = -\frac{x_2}{-8} = \frac{x_3}{-16}$$

$$\frac{x_1}{2} = \frac{x_2}{1} = \frac{x_3}{-2} = k_2, (k_2 \neq 0)$$

$\therefore x_1 = 2k_2, x_2 = k_2$ and $x_3 = -2k_2$. Substitute x_1, x_2 and x_3 in the eigenvector X .
 $X_2 = [2k_2, k_2, -2k_2]^T$, if $k_2 = 1$, then $X_2 = [2, 1, -2]^T$ is eigenvector corresponding to the eigenvalue $\lambda = 3$ of the given matrix A .

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Case - 03: Substituting $\lambda = 15$ in the above matrix equation (1), we get

$$-7x_1 - 6x_2 + 2x_3 = 0$$

$$-6x_1 - 8x_2 - 4x_3 = 0$$

$$2x_1 - 4x_2 - 12x_3 = 0$$

solve by using cross multiplication method

$$\frac{x_1}{(24 + 16)} = -\frac{x_2}{(28 + 12)} = \frac{x_3}{(56 - 36)}$$

$$\frac{x_1}{40} = -\frac{x_2}{40} = \frac{x_3}{20}$$

$$\frac{x_1}{2} = \frac{x_2}{-2} = \frac{x_3}{1} = k_3, (k_3 \neq 0)$$

$\therefore x_1 = 2k_3, x_2 = -2k_3$ and $x_3 = k_3$. Substitute x_1, x_2 and x_3 in the eigenvector X .
 $X_3 = [2k_3, -2k_3, k_3]^T$, if $k_3 = 1$, then $X_3 = [2, -2, 1]^T$ is eigenvector corresponding to the eigenvalue $\lambda = 15$ of the given matrix A .

Continued Unit - IV Problems on Eigenvalues and Eigenvector of a Matrix ...

Problems: Find all the eigenvalues and corresponding eigenvectors of the following matrices.

$$(1) \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix} \quad (2) \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \quad (3) \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

Continued Unit - IV on Rayleigh's Power Method (OR) Power Method

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Rayleigh's Power Method (OR) Power Method:

Rayleigh's Power Method is a numerically an iterative method to obtain an approximate the largest eigenvalue and corresponding eigenvector of the given matrix A . This method is useful to write the programming for find the eigenvalues and corresponding eigenvectors of the matrix A .

Let $X^{(0)}$ be an initial eigenvector and A be the given matrix, then

Iteration - 1:

$$AX^{(0)} = \lambda^{(1)} X^{(1)}$$

Iteration - 2:

$$AX^{(1)} = \lambda^{(2)} X^{(2)}$$

Iteration - 3:

$$AX^{(2)} = \lambda^{(3)} X^{(3)}$$

Iteration - 4:

$$AX^{(3)} = \lambda^{(4)} X^{(4)}$$

Iteration - 5:

$$AX^{(4)} = \lambda^{(5)} X^{(5)}$$

Iteration - 6:

$$AX^{(5)} = \lambda^{(6)} X^{(6)}$$

Continued Unit - IV on Rayleigh's Power Method (OR) Power Method

...

Continue this iteration process till two consecutive iterative values of λ and X are same at least two decimal places. The value so obtained at last iteration values are respectively the largest eigenvalue and the corresponding eigenvector of the given matrix A .

Note: Suppose an initial eigenvector $X^{(0)}$ is not given in the problem, choose an initial eigenvector like $X^{(0)} = [1, 0, 0]^T$ (or) $X^{(0)} = [0, 1, 0]^T$ (or) $X^{(0)} = [0, 0, 1]^T$ (or) $X^{(0)} = [1, 1, 1]^T$ for the matrix of order 3.

Continued Unit - IV Problems on Rayleigh's Power Method (OR) Power Method ...

Problems:

Problem - 01: Find the largest eigenvalue and the corresponding eigenvector of the matrix A by using Power method.

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Solution: Given

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Now, taking $X^{(0)} = [1, 0, 0]^T$ be the initial eigenvector

Iteration - 1:

$$AX^{(0)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \lambda^{(1)} X^{(1)}$$

Iteration - 2:

$$AX^{(1)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 0 \\ 2 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \lambda^{(2)} X^{(2)}$$

Iteration - 3:

$$AX^{(2)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 2.8 \\ 0 \\ 2.6 \end{bmatrix} = 2.8 \begin{bmatrix} 1 \\ 0 \\ 0.93 \end{bmatrix} = \lambda^{(3)} X^{(3)}$$

Iteration - 4:

$$AX^{(3)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.93 \end{bmatrix} = \begin{bmatrix} 2.93 \\ 0 \\ 2.86 \end{bmatrix} = 2.93 \begin{bmatrix} 1 \\ 0 \\ 0.98 \end{bmatrix} = \lambda^{(4)} X^{(4)}$$

Continued Unit - IV Problems on Rayleigh Power Method ...

Iteration - 5:

$$AX^{(4)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.98 \end{bmatrix} = \begin{bmatrix} 2.98 \\ 0 \\ 2.96 \end{bmatrix} = 2.98 \begin{bmatrix} 1 \\ 0 \\ 0.99 \end{bmatrix} = \lambda^{(5)} X^{(5)}$$

Iteration - 6:

$$AX^{(5)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.99 \end{bmatrix} = \begin{bmatrix} 2.99 \\ 0 \\ 2.98 \end{bmatrix} = 2.99 \begin{bmatrix} 1 \\ 0 \\ 0.997 \end{bmatrix} = \lambda^{(6)} X^{(6)}$$

Iteration - 7:

$$AX^{(6)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.997 \end{bmatrix} = \begin{bmatrix} 2.997 \\ 0 \\ 2.994 \end{bmatrix} = 2.997 \begin{bmatrix} 1 \\ 0 \\ 0.999 \end{bmatrix} = \lambda^{(7)} X^{(7)}$$

Therefore, we can observe that the values of $\lambda^{(6)} \approx \lambda^{(7)}$ and $x^{(6)} \approx x^{(7)}$. Now, stop the iteration process and hence $\lambda = 2.997 \approx 3$ is the largest eigenvalue and the corresponding eigenvector $X = [1, 0, 0.999]^T \approx [1, 0, 1]^T$ of the given matrix A .

Continued Unit - IV Problems on Rayleigh Power Method ...

Problem - 02: Apply Rayleigh's power method to find the dominant eigenvalue and the corresponding eigenvector of the matrix A , by taking the initial eigenvector as $[1, 1, 1]^T$.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Solution: By data

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

and initial eigenvector $X^{(0)} = [1, 1, 1]^T$

Iteration - 1:

$$AX^{(0)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 4 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 0 \\ 0.67 \end{bmatrix} = \lambda^{(1)} X^{(1)}$$

Iteration - 2:

$$AX^{(1)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.67 \end{bmatrix} = \begin{bmatrix} 7.34 \\ -2.67 \\ 4.01 \end{bmatrix} = 7.34 \begin{bmatrix} 1 \\ -0.36 \\ 0.55 \end{bmatrix} = \lambda^{(2)} X^{(2)}$$

Iteration - 3:

$$AX^{(2)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.36 \\ 0.55 \end{bmatrix} = \begin{bmatrix} 7.82 \\ -3.63 \\ 4.01 \end{bmatrix} = 7.82 \begin{bmatrix} 1 \\ -0.46 \\ 0.51 \end{bmatrix} = \lambda^{(3)} x^{(3)}$$

Iteration - 4:

$$AX^{(3)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.46 \\ 0.51 \end{bmatrix} = \begin{bmatrix} 7.94 \\ -3.89 \\ 3.99 \end{bmatrix} = 7.94 \begin{bmatrix} 1 \\ -0.49 \\ 0.5 \end{bmatrix} = \lambda^{(4)} x^{(4)}$$

Continued Unit - IV Problems on Rayleigh Power Method ...

Iteration - 5:

$$AX^{(4)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.49 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 7.98 \\ -3.97 \\ 3.99 \end{bmatrix} = 7.98 \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \lambda^{(5)} x^{(5)}$$

Iteration - 6:

$$AX^{(4)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 8 \\ -4 \\ 4 \end{bmatrix} = 8 \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \lambda^{(6)} x^{(6)}$$

Therefore, we can observe that the values of $\lambda^{(5)} \approx \lambda^{(6)}$ and $x^{(5)} \approx x^{(6)}$. Now, stop the iteration process and hence $\lambda = 8$ is the largest eigenvalue and the corresponding eigenvector $X = [1, -0.5, 0.5]^T$ of the given matrix A . **Problem - 03:** Apply Rayleigh's power method to find the dominant eigenvalue and the corresponding eigenvector of the matrix A .

$$A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$$

Continued Unit - IV Problems on Rayleigh Power Method ...

Solution: By data

$$A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$$

and taking an initial eigenvector $X^{(0)} = [1, 0, 0]^T$

Iteration - 1:

$$AX^{(0)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 25 \\ 1 \\ 2 \end{bmatrix} = 25 \begin{bmatrix} 1 \\ 0.04 \\ 0.08 \end{bmatrix} = \lambda^{(1)} X^{(1)}$$

Iteration - 2:

$$AX^{(1)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0.04 \\ 0.08 \end{bmatrix} = \begin{bmatrix} 25.2 \\ 1.12 \\ 1.68 \end{bmatrix} = 25.2 \begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix} = \lambda^{(2)} X^{(2)}$$

Iteration - 3:

$$\begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$$

Problems: Applying Power method to obtain a largest eigenvalue and corresponding eigenvector of the following matrices.

$$(1) \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

by taking $[0, 1, 0]^T$ as initial eigenvector.

$$(2) \begin{bmatrix} 4 & 1 & 0 \\ 1 & 20 & 1 \\ 0 & 1 & 4 \end{bmatrix} \quad (3) \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

Continued Unit - IV on Diagonalization of a Square Matrix ...

Singular matrix: A square matrix A of order n is said to be a **singular matrix** if $|A| = 0$.

Non - singular matrix: A square matrix A of order n is said to be a **Non - singular matrix** if $|A| \neq 0$.

Similarity Transformation: If A and B are any two square matrices of order n , if there exist a non - singular matrix P such that $P^{-1}AP = B$, then the matrix A is said to be similar to the matrix B .

Similarly, if $P^{-1}BP = A$, then the matrix B is said to be similar to the matrix A .

Diagonalization of a square matrix: Suppose A be any given square matrix of order n . The process of reducing a given square matrix A to diagonal matrix is known as a diagonalization of a square matrix A .

Suppose A be any given square matrix of order n , if there exist a non - singular matrix P such that $P^{-1}AP = D$, OR $AP = PD$ then the matrix A is said to be a diagonalizable matrix.

$$P^{-1}AP = D$$

OR

$$AP = PD$$

Model matrix: Let A be any given square matrix of order n , if there exist a non - singular matrix P such that $P^{-1}AP = D$, then a non - singular matrix P is called as a

Continued Unit - IV on Diagonalization of a Square Matrix ...

Spectral matrix: Let A be any given square matrix of order n , if there exist a non-singular matrix P such that $P^{-1}AP = D$, then a diagonal matrix D is called as a **Spectral matrix** whose diagonal elements are the eigenvalues of the given square matrix A and it is given by,

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & 0 & \dots & 0 \\ 0 & 0 & \lambda_4 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Power of a matrix: Let A be any given square matrix of order n , then the process of finding the powers of the given matrix A using the concept of a diagonalization of a square matrix is known as a power of a square matrix.

We know that, by diagonalization of a matrix A is

$$P^{-1}AP = D$$

pre multiply by P and post multiply by P^{-1} on both sides, we get

$$(P)(P^{-1}AP)(P^{-1}) = (P)D(P^{-1})$$

Continued Unit - IV on Diagonalization of a Square Matrix ...

$$(PP^{-1})(A)(PP^{-1}) = PDP^{-1}$$

but, $PP^{-1} = I$ and $AI = IA = A$, $IAI = A$

$$(I)(A)(I) = PDP^{-1}$$

$$(IAI) = PDP^{-1}$$

$$A = PDP^{-1}$$

Therefore, consider the matrix A^2

$$A^2 = (A)(A) = (PDP^{-1})(PDP^{-1})$$

$$A^2 = (PD)(P^{-1}P)(DP^{-1})$$

$$A^2 = (PD)(I)(DP^{-1}) = (PD)(DP^{-1})$$

$$A^2 = PD^2P^{-1}$$

similarly, continue like this we get

$$A^3 = PD^3P^{-1}$$

$$A^4 = PD^4P^{-1}$$

In general, for a positive integer m

$$A^m = PD^mP^{-1}$$

Continued Unit - IV on Diagonalization of a Square Matrix ...

Working procedure to reducing the given matrix to diagonal matrix:

Step - 01: Write the characteristics equation for the given matrix A of the form $|A - \lambda I| = 0$, expanding this determinant find all eigenvalues of the given matrix A .

Step - 02: Obtain all eigenvectors corresponding each eigenvalues of the given matrix A by taking eigenvector of the form $X = [x, y, z]^T$ and using matrix equation $[A - \lambda I][X] = [0]$.

Step - 03: Verify the given matrix A is reducing to a diagonal matrix by using the formula for diagonalization of a matrix .

$$P^{-1}AP = D$$

where

$$P^{-1} = \frac{1}{|P|} \text{Adj}(P)$$

OR

$$AP = PD$$

Step - 04: Find the power of the given matrix A by using the formula .

$$A^m = PD^m P^{-1}$$

Continued Unit - IV on Diagonalization of a Square Matrix ...

$$D^m = \begin{bmatrix} \lambda_1^m & 0 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2^m & 0 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3^m & 0 & \dots & 0 \\ 0 & 0 & \lambda_4^m & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \lambda_m^m \end{bmatrix}$$

Problem - 01: Diagonalize the matrix and hence find A^6 .

$$\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$$

Solution: Given a matrix

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$$

We know that, the characteristics equation

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2 - \lambda & 2 \\ 1 & 3 - \lambda \end{vmatrix} = 0$$

Now, expanding the determinant, we get, the eigen values are 1 and 4

Continued Unit - IV Problems on Diagonalization of a Square Matrix ...

Let $X = [x, y]^T$ be the eigen vector and matrix equation is

$$[A - \lambda I][X] = [0]$$

$$\begin{bmatrix} 2 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (34)$$

Case - 01: Substituting $\lambda = 1$ in the above matrix equation (31), we get

$$x + 2y = 0$$

and

$$x + 2y = 0$$

both are same

$$x = -2y$$

$$\frac{x}{2} = \frac{y}{-1} = k_1, (k_1 \neq 0)$$

$$\frac{x}{2} = k_1, x = 2k_1, \frac{y}{-1} = k_1, y = -k_1$$

Substitute $x = 2k_1$ and $y = -k_1$ in the eigen vector X , $X_1 = [2k_1, -k_1]^T$ and $X_1 = [2, -1]^T$, if $(k_1 = 1)$ be the eigen vector corresponding to the eigen value $\lambda_1 = 1$.

Continued Unit - IV Problems on Diagonalization of a Square Matrix ...

Case - 02: Substituting $\lambda = 4$ in the above matrix equation (31), we get

$$-2x + 2y = 0$$

and

$$x - y = 0$$

both are same

$$x = y$$

$$\frac{x}{1} = \frac{y}{1} = k_2, (k_2 \neq 0)$$

$$\frac{x}{1} = k_2, x = k_2, \frac{y}{1} = k_2, y = k_2$$

Substitute $x = k_2$ and $y = k_2$ in the eigen vector X , $X_2 = [k_2, k_2]^T$ and $X_2 = [1, 1]^T$, if $(k_2 = 1)$ be the eigen vector corresponding to the eigen value $\lambda_1 = 4$.

$\therefore$ Model matrix

$$P = [X_1, X_2]$$

$$P = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$$

Spectral matrix D is

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

We know that, by diagonalization of a matrix A is

$$P^{-1}AP = D$$

$$P^{-1}AP = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

To find A^6 : We have,

$$A^6 = PD^6P^{-1}$$

$$A^6 = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}^6 \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$$

$$A^6 = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1^6 & 0 \\ 0 & 4^6 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}^{-1}$$

$$A^6 = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4096 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1366 & 2730 \\ 1365 & 2731 \end{bmatrix}$$

Problem - 02: Diagonalize the matrix and hence find A^6 .

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

Solution: Given a matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

Continued Unit - IV on Application Linear Algebra to solve the system of a first order and first degree ODE's using Matrix Method ...

Consider a simple first order and first degree ODE of the form

$$\frac{dy}{dx} = ay$$

where a is any constant and y is unknown function to be determine. Find the general solution of the ODE

Method - I: Using the method of separation of variables

$$\frac{dy}{y} = a dx$$

Integrating on both sides, we get

$$\int \frac{dy}{y} = a \int dx + c$$

$$\log(y) = ax + c$$

$$y = e^{ax+c} = e^{ax} e^c$$

Continued Unit - IV on Application of Linear Algebra to solve the system of a first order and first degree ODE's using Matrix Method ...

$$y = c_1 e^{ax}, c_1 = e^c$$

Therefore, $y = c_1 e^{ax}$ is the general solution of the above ODE.

Method - II: Using the method to solve second and higher order LDE

$$\frac{dy}{dx} = ay$$

$$Dy = ay, D = \frac{d}{dx}$$

$$Dy - ay = 0$$

$$(D - a)y = 0$$

The A. E. is $f(m) = 0$, $m - a = 0$, $m = a$, C.F. $= c_1 e^{ax}$ and P.I. $= 0$, because RHS is zero. the general solution is $y = c_1 e^{ax}$ of the above ODE.

Note: suppose $\frac{dy}{dx} = ay$ is the given first order and first degree ordinary differential equation then the general solution is $y = c_1 e^{ax}$.

Continued Unit - IV on Application of Linear Algebra to solve the system of a first order and first degree ODE's using Matrix Method ...

Consider a set of m first order and first degree ordinary differential equation in n unknowns as follows

$$y_1' = a_{11}y_1 + a_{12}y_2 + a_{13}y_3 + \dots + a_{1n}y_n$$

$$y_2' = a_{21}y_1 + a_{22}y_2 + a_{23}y_3 + \dots + a_{2n}y_n$$

$$y_3' = a_{31}y_1 + a_{32}y_2 + a_{33}y_3 + \dots + a_{3n}y_n$$

...

...

$$y_n' = a_{m1}y_1 + a_{m2}y_2 + a_{m3}y_3 + \dots + a_{mn}y_n$$

Where, $y_1 = f_1(x)$, $y_2 = f_2(x)$, $y_3 = f_3(x)$, ..., $y_n = f_n(x)$ are the unknown functions to be determine.

The above system of ODE's in matrix equation is

$$y' = Ay$$

where, A is the coefficient matrix of the given system of ODE.

Continued Unit - IV on Application of Linear Algebra to solve the system of a first order and first degree ODE's using Matrix Method ...

$$\begin{bmatrix} y_1' \\ y_2' \\ y_3' \\ \dots \\ y_n' \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \dots \\ y_n \end{bmatrix}$$

where, the coefficient matrix A is

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

Continued Unit - IV Problems on Application of Linear Algebra to solve the system of a first order and first degree ODE's using Matrix Method

...

Problem - 1: Write the following system of ODE's in matrix form and hence solve using matrix method

$$y_1' = -2y_1, y_2' = 3y_2, y_3' = y_3$$

given that $y_1(0) = 1$, $y_2(0) = -1$ and $y_3(0) = 2$.

Solution: The given system of ODE's

$$y_1' = -2y_1, y_2' = 3y_2, y_3' = y_3$$

and initial conditions $y_1(0) = 1$, $y_2(0) = -1$ and $y_3(0) = 2$.

The given system of ODE's in matrix equation

$$y' = Ay$$

$$\begin{bmatrix} y_1' \\ y_2' \\ y_3' \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Continued Unit - IV Problems on Application of Linear Algebra to solve the system of a first order and first degree ODE's using Matrix Method

...

The general solutions for the given system of ODE's are

$$y_1' = -2y_1 \rightarrow y_1 = c_1 e^{-2x}$$

$$y_2' = 3y_2 \rightarrow y_2 = c_2 e^{3x}$$

$$y_3' = y_3 \rightarrow y_3 = c_3 e^x$$

$\therefore y_1 = c_1 e^{-2x}$, $y_2 = c_2 e^{3x}$ and $y_3 = c_3 e^x$ are the general solutions for the given system of ODE's. Also given the initial conditions

$$y_1(0) = 1, \text{ when } x = 0, y_1 = 1$$

$$y_1 = c_1 e^{-2x} \rightarrow c_1 = 1, y_1 = e^{-2x}$$

$$y_2(0) = -1, \text{ when } x = 0, y_2 = -1$$

$$y_2 = c_2 e^{3x} \rightarrow c_2 = -1, y_2 = -e^{3x}$$

$$y_3(0) = 2, \text{ when } x = 0, y_3 = 2$$

$$y_3 = c_3 e^x \rightarrow c_3 = 2, y_3 = 2e^x$$

Hence $y_1 = e^{-2x}$, $y_2 = -e^{3x}$ and $y_3 = 2e^x$ are the particular solutions for the given system of ODE's

Continued Unit - IV Problems on Application of Linear Algebra to solve the system of a first order and first degree ODE's using Matrix Method

...

Note: In the above examples, in the given system of first order and first degree ODE's the coefficient matrix A is diagonal matrix and hence it is easy to solve the given system of first order and first degree ODE's. If the coefficient matrix A of the given system of a first order and first degree ODE's is not in diagonal matrix then reduce it to diagonal matrix using the concept of the diagonalization of the square matrix and solve the given system of a first order and first degree ODE's by taking the substitution $y = Pu$ and $u' = Du$, where P is the modal matrix whose columns are eigenvectors of the coefficient matrix A and D is the spectral matrix whose diagonal elements are eigenvalues of the coefficient matrix A of the given system of a first order and first degree ODE's.

$$y = pu$$
$$y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

The matrix P is a modal matrix of the coefficient matrix A and is given by

$$P = [X_1, X_2, X_3]$$

Continued Unit - IV Problems on Application of Linear Algebra to solve the system of a first order and first degree ODE's using Matrix Method

...

$$u' = Du$$
$$u' = \begin{bmatrix} u_1' \\ u_2' \\ u_3' \end{bmatrix}, u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

The diagonal matrix D is a spectral matrix of the coefficient matrix A and it is given by

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

Continued Unit - IV Problems on Application of Linear Algebra to solve the system of a first order and first degree ODE's using Matrix Method

...

Problem - 2: Write the following system of ODE's in matrix form and hence solve using matrix method

$$x_1' = x_1 + x_2, x_2' = 3x_1 + 3x_2$$

given that $x_1(0) = 0, x_2(0) = 0$.

Problem - 3: Write the following system of ODE's in matrix form and hence solve using matrix method

$$y_1' = y_1 + y_2, y_2' = 4y_1 - 2y_2$$

given that $y_1(0) = 1, y_2(0) = 6$.

Problem - 4: Write the following system of ODE's in matrix form and hence solve using matrix method

$$y_1' = 4y_1 + 2y_2 + 2y_3, y_2' = 2y_1 + 4y_2 + 2y_3, y_3' = 2y_1 + 2y_2 + 4y_3$$